

ADITYA DINESH RAJPUT

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Embedded Systems | Linux | Robotics | Web Development

EDUCATION

PDPM IIITDM Jabalpur

B.Tech in Electronics and Communication Engineering

CGPA: 9.3

2024 - 2028

TECHNICAL SKILLS

Languages: C, C++, Python, Rust, Java, JavaScript, Lua

Systems & Embedded: Linux, STM32, UART, SPI, I2C, RTOS, ARM Assembly

Tools: Git, GDB, QEMU, STM32CubeIDE, NeoVim, MATLAB

Frameworks: TensorFlow, PyTorch, React

PROJECTS

Real Time Operating System (RTOS) | C, RTOS, ARM Assembly

GitHub Repo

- Implemented a small RTOS from scratch in C.
- It uses Preemptive Scheduling for scheduling tasks.
- Implemented the context-switcher in ARM Assembly.
- It has many features like Queues, Semaphores, and Mutexes.
- Debugged and tested using GDB.

STM32F411 Bare-Metal Device Drivers | Embedded C, STM32CubeIDE

GitHub Repo

- Implemented bare-metal register-level drivers for GPIO, UART, SPI, and I2C drivers for STM32F411 without using HAL libraries.
- Configured peripheral clocks, memory-mapped registers, and communication protocols directly using the STM32 reference manual.
- Debugged baud-rate and peripheral clock issues during UART driver development.

Self Balancing Bot Simulation | CoppeliaSim, Python

- Designed and simulated a self-balancing robot in CoppeliaSim using PID-based stabilization and real-time sensor feedback.
- Fine-tuned PID controller parameters to improve balance stability and response time.

VCG Signal Compression Using Machine Learning | TensorFlow, Signal Processing

GitHub Repo

- Developed a CNN-LSTM autoencoder model for VCG signal compression and reconstruction.
- Trained and evaluated the model using TensorFlow-based workflows for biomedical signal processing.

Chess Vision System | Python, OpenCV, YOLO, Stockfish

GitHub Repo

- Developed a computer vision pipeline that detects chess pieces and reconstructs board state using YOLO-based object detection.
- Generated FEN strings from detected board positions and integrated Stockfish to suggest optimal moves.
- Implemented board warping and grid mapping using OpenCV for accurate position tracking.

EXPERIENCE

E-Yantra 2025

IIT Bombay

Reached Simulation Stage

- Developed a simulation for a self-balancing robot in CoppeliaSim.

Techkriti 2026 - IARC Competition

IIT Kanpur

1st Place

- Developed and optimized an autonomous maze-solving robot that achieved the fastest completion time in the competition.

LEADERSHIP & EXTRACURRICULAR

Electronics and Robotics Society

IIITDM Jabalpur

Core Committee Member

- Organized workshops and events focused on embedded systems, robotics, and electronics for students.
- Mentored first-year students during RoboRush 4.0 in building WiFi-controlled robots.